SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I M MD Rehan (192425391) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MMDRehan

Scenario-Based Questions

CO-AT2

QUESTIONS: 05 Total Marks:25

1. **Unsupervised Training of Neural Networks, Auto Encoders**
   A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. **Auto Encoders, Representation Learning**

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

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1. Unsupervised Training of Neural Networks – ECG Abnormality Detection

Recommended Technique: Deep Autoencoder for Anomaly Detection

In the given scenario, where a hospital needs to analyze thousands of **unlabeled ECG records**, a **Deep Autoencoder** is the most appropriate technique. Since labeled medical datasets are often scarce and expensive to obtain, unsupervised learning becomes essential. Autoencoders are specifically designed to learn the **intrinsic structure and patterns** of input data without requiring labels, making them highly suitable for medical signal analysis.

An autoencoder learns to reconstruct its input by passing it through a compressed latent space. When trained predominantly on **normal ECG signals**, the network captures the characteristic patterns of healthy heart activity. However, when abnormal signals are introduced, the model fails to reconstruct them accurately, leading to a **high reconstruction error**, which serves as an indicator of anomaly.

Justification of Choice

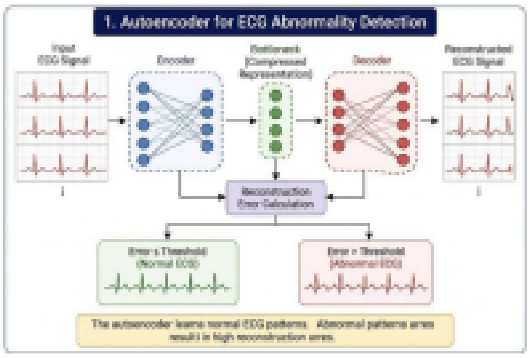
- Handles **unlabeled datasets** efficiently
- Learns **compact and meaningful representations**
- Detects subtle deviations in ECG waveform patterns
- Scalable for large healthcare systems

Workflow of the Proposed System

The implementation begins with preprocessing ECG signals, including normalization and noise filtering to ensure data consistency. The processed signals are then passed through the autoencoder network, where the encoder compresses the data into a lower-dimensional representation. The decoder reconstructs the signal, and the reconstruction error is computed. A predefined threshold is used to classify signals as normal or abnormal. This approach enables continuous monitoring and early detection of cardiac irregularities.

Key Advantages

- Eliminates dependency on labeled data
- Effective for rare and unseen abnormalities
- Provides automated and real-time analysis



Autoencoder Architecture: Input → Encoder → Bottleneck → Decoder → Out

2. Autoencoders for Dimensionality Reduction in Retail Data

In retail analytics, customer purchase data is often **high-dimensional and complex**, containing numerous variables such as purchase frequency, product categories, and spending patterns. Directly applying clustering algorithms on such data can lead to inefficiencies and poor segmentation. To address this, **Autoencoders can be effectively used for dimensionality reduction**, enabling the extraction of meaningful and compact feature representations.

Unlike traditional techniques such as PCA, autoencoders capture **non-linear relationships** within the data, making them more suitable for modeling real-world consumer behavior. The encoder compresses the input data into a lower-dimensional latent space, while the decoder ensures that essential information is retained during reconstruction.

Working Mechanism

- Input data is passed through multiple hidden layers
- Encoder reduces dimensionality
- Bottleneck layer stores compressed features
- Decoder reconstructs original data

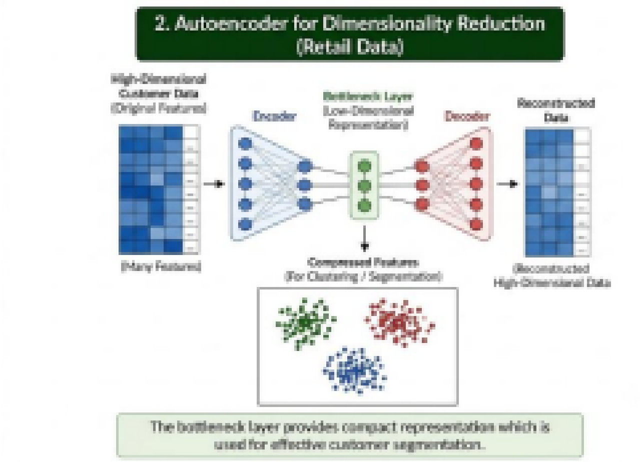
Application in Customer Segmentation

The compressed representation obtained from the bottleneck layer can be directly used as input for clustering algorithms such as K-Means. This improves segmentation quality by removing redundant and irrelevant features while preserving important purchasing patterns.

Benefits

- Reduces computational complexity
- Enhances clustering performance
- Captures hidden patterns in customer behavior
- More powerful than linear techniques like PCA

High-Dimensional Data → Encoder → Bottleneck → Reduced Features



3. Improving Facial Recognition using Network Depth and Activation Functions

Facial recognition systems require the ability to learn highly complex and hierarchical features from image data. A **shallow neural network** lacks sufficient capacity to capture such intricate patterns, resulting in poor performance. To overcome this limitation, it is essential to redesign the architecture by increasing network depth and selecting appropriate activation functions.

Increasing the depth of the network allows the model to learn **multi-level feature representations**, starting from simple edges in initial layers to complex facial structures in deeper layers. This hierarchical learning significantly improves recognition accuracy.

Architectural Improvements

1. Increase Network Depth

- Enables hierarchical feature extraction
- Improves representation learning capability
- Captures complex facial variations

2. Use Advanced Activation Functions

- ReLU: Faster training and avoids vanishing gradient
- Leaky ReLU: Prevents dead neuron problem
- Softmax: Suitable for classification output

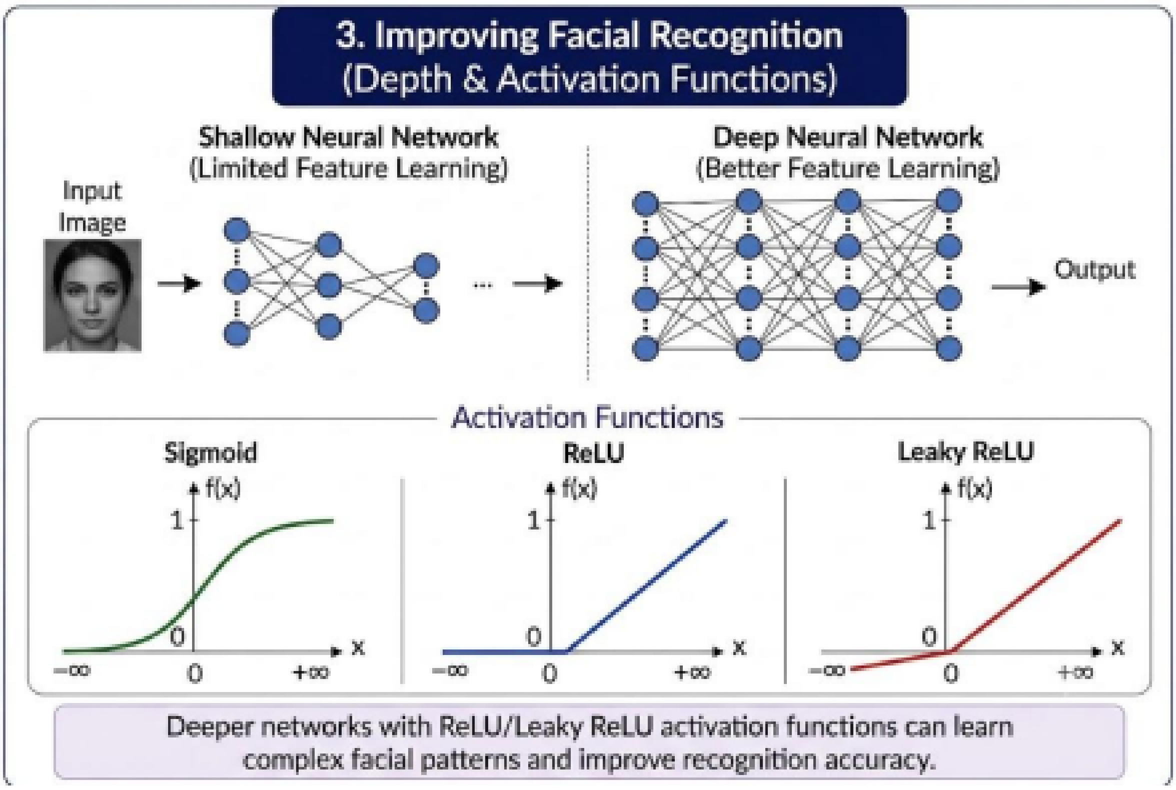
Additional Enhancements

- Use Convolutional Neural Networks (CNNs) for image data
- Apply Batch Normalization for stability
- Include Dropout layers to reduce overfitting

Summary Points

- Shallow networks → Limited learning capability
- Deep networks → Better feature abstraction
- Proper activation → Efficient training

Comparison of Shallow vs Deep Neural Network



4. Movie Recommendation using Restricted Boltzmann Machines (RBMs)

For a streaming service that aims to recommend movies without labeled data, **Restricted Boltzmann Machines (RBMs)** provide an effective unsupervised learning solution. RBMs are probabilistic neural networks capable of learning hidden patterns in user preferences by modeling the relationships between users and items.

An RBM consists of a **visible layer**, representing user interactions (such as ratings or watch history), and a **hidden layer**, which captures latent features such as user interests and preferences. The model learns by adjusting weights to minimize the difference between observed and reconstructed data.

Application in Recommendation Systems

RBMs analyze the user-item interaction matrix and learn underlying patterns in viewing behavior. By reconstructing missing values in the matrix, the model predicts user preferences for unseen movies, enabling personalized recommendations.

Process Flow

- Construct user-movie interaction matrix
- Train RBM using unsupervised learning
- Learn hidden feature representations
- Predict missing ratings
- Recommend top-rated movies

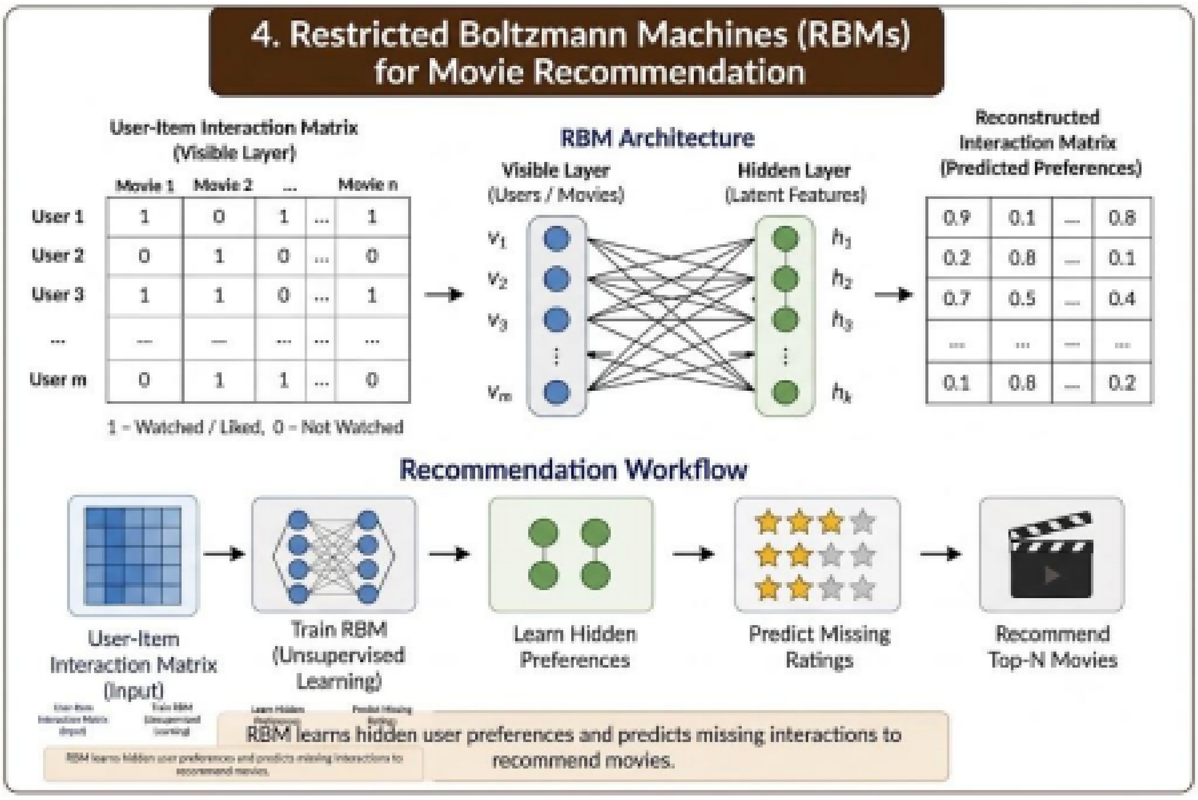
Advantages

- Works without labeled data
- Captures complex user preferences
- Handles sparse datasets effectively

Key Characteristics

- Energy-based model
- Probabilistic learning approach
- Suitable for collaborative filtering

RBM Structure: Visible Layer ↔ Hidden Layer



5. Defective Product Detection using Deep Learning

In a manufacturing environment where sensor data is continuously collected, detecting defective products requires an intelligent system capable of identifying subtle deviations in patterns. A **deep learning-based solution combining representation learning and neural networks** is highly effective for this purpose.

The approach involves using an **Autoencoder for feature extraction** followed by a **Neural Network for classification**. The autoencoder learns meaningful representations from raw sensor data, capturing essential patterns while eliminating noise. These learned features are then used by a classifier to distinguish between defective and non-defective products.

Proposed Approach

The system begins with preprocessing sensor data, including normalization and noise removal. The autoencoder is trained to learn compact feature representations. These features are then fed into a supervised neural network that performs classification based on learned patterns.

Model Structure

- Input Layer: Sensor readings
- Hidden Layers: Feature extraction and learning
- Output Layer: Defective / Non-defective classification

Advantages

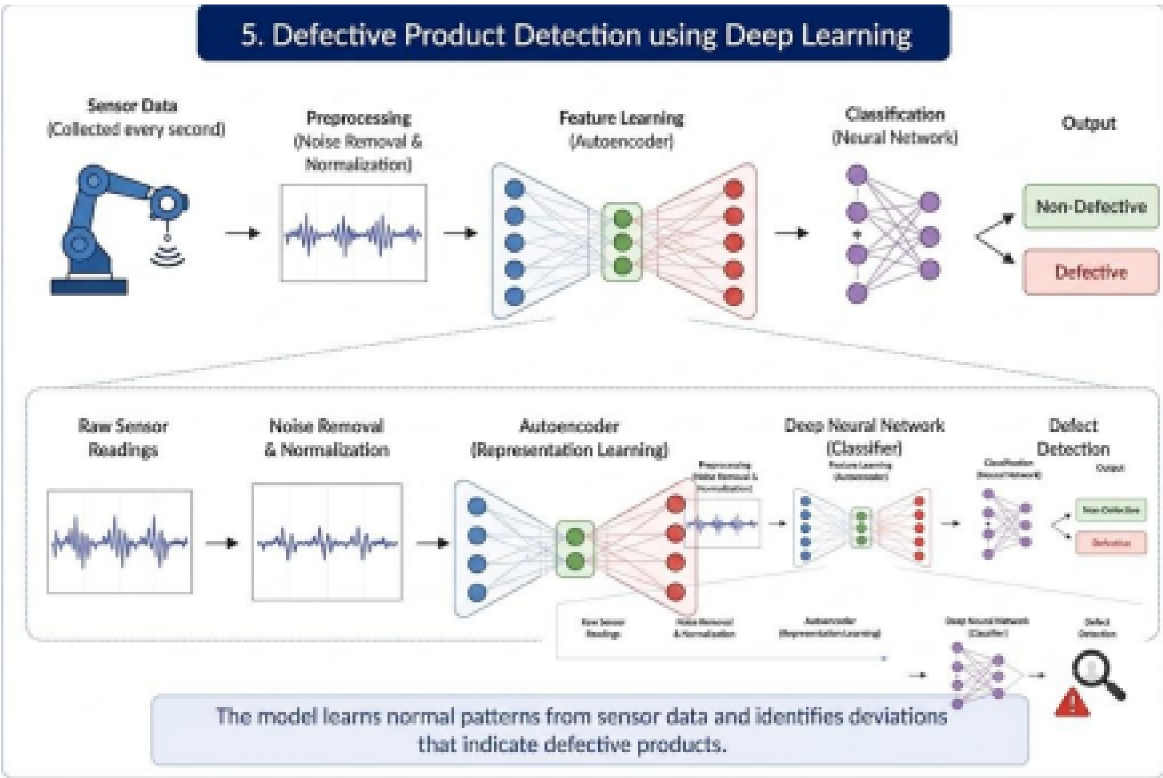
- Enables real-time defect detection
- Reduces manual inspection efforts
- Improves production quality and efficiency

Key Points

- Representation learning extracts meaningful features
- Neural networks perform accurate classification
- Suitable for large-scale industrial data

(Leave Space for Diagram)

Pipeline: Sensor Data → Feature Learning → Classification



Conclusion

The above scenarios collectively demonstrate the significance of supervised and unsupervised deep learning techniques in solving complex real-world problems across multiple domains such as healthcare, retail, entertainment, and manufacturing. Techniques like Autoencoders and Restricted Boltzmann Machines enable effective representation learning, allowing systems to extract meaningful patterns from high-dimensional and unlabeled data. Furthermore, architectural improvements such as deeper neural networks and optimized activation functions enhance model performance in tasks like image recognition and classification. The integration of these approaches not only improves prediction accuracy but also ensures scalability, efficiency, and automation in decision-making processes. Overall, the ability to select, train, and evaluate appropriate models reflects the practical application of Course Outcome CO2 in addressing real-world analytical challenges.

• Deep learning models effectively handle both **labeled and unlabeled data scenarios** • Representation learning helps in extracting **meaningful and compact features** • Model performance depends on **architecture design and activation functions**

• Unsupervised techniques like Autoencoders and RBMs are crucial for **real-world applications**

• Proper model selection and evaluation ensure **accurate and scalable solutions**.